

Final Year Project - Mid Report



“FearLESS” Development

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Introduction

The purpose of this mid evaluation report is to define the scope, system level requirements, and first iteration deliverables for our VR based application named “FearLESS”. FearLESS helps people overcome their fears of heights (Acrophobia) and darkness (Nyctophobia) using VR, gauges a person’s fear intensity, and connects its users to therapists/psychiatrists for counseling. This document is intended for the FearLESS team, our supervisor Mr. Bilal Khalid Dar, our co-supervisor Mr. Wasif Ali Wasif, and the FYP panel.

Problem Statement

Fear is a natural survival response to danger that becomes irrational, dysfunctional, and even phobic when the level of danger is insignificant compared to the reaction it caused [1]. Such irrational fears and/or phobias vary in intensity amongst individuals. They often disturb the suffering individual’s daily life, making simple daily activities and pleasures difficult to achieve, as they go out of their way to overcome their fear.

For instance, an irrational fear of heights would involve the person having panic attacks by standing on a ladder or walking up a flight of stairs. A person’s fear of darkness may cause them to miss sleep, depression, and they limit their life to the daytime since leaving the house after dark becomes an extremely stressful and difficult task. Avoidance strategies do help the person avoid their fears and triggers, but it can also limit what a person can do in their life, and may even become dangerous for the person. An irrational fear of clowns may cause the person to miss their family’s birthday parties and celebrations. Likewise, a person can miss medical appointments, and life saving drugs due to their intense fear of needles.

According to the National Comorbidity Survey, among the adults with phobias in the United States, “an estimated 21.9% had high fear intensity, 30.0% had moderate impairment, and 48.1% had mild impairment” [3]. Treatment for such fears, and phobias include exposure therapy and cognitive-behavioral therapy (CBT), however, only about 23% of the people with phobias seek treatment [2]. Taking therapy or even checking the intensity of one’s fear requires funds and access to professionals which most people end up avoiding, either because of their geographic location, or social stigma that comes with seeking help from mental health professionals. Moreover, it is a challenge to control and replicate the exposure therapy treatment in a safe environment.

Problem Solution

The proposed solution to the above problem statement is FearLESS, an application that uses Virtual Reality Exposure Therapy (VRET) to help users confirm and overcome their irrational fears/phobias. With the help of FearLESS, users can expose themselves to their fears, using simulations, at varying intensities, until they can conquer and control their reactions to their irrational fears/phobias.

To confirm the presence of irrational fear/phobia, and to determine the severity of the fear, FearLESS allows users to take a questionnaire before they can view the simulations. The results of the

questionnaire are used by FearLESS to either adjust the intensity of the fear shown in the simulation, or to suggest psychiatrists and therapists, available on FearLESS, to guide, and help the users before they view the simulations. This is done for the user's mental safety. Furthermore, users can also use these questionnaires to track their progress, as they face their fears in the simulations.

The fears that are offered by the application include **Acrophobia** (fear of heights) [4], and **Nyctophobia** (fear of darkness) [8]. The simulations for these fears are provided to the users in three different levels (increasing intensity). The level users can view (as many times they need) is based on the questionnaire score. Therapists can access the simulations to provide exposure therapy from the comfort of their own office/clinic.

Modules

Our project includes 8 different modules, which are as follows:

- 1. Phobia Selection:**

This module allows users to view exposure therapy simulations for different fears/phobias. FearLESS will offer simulations for two fears, namely Acrophobia and Nyctophobia.

- 2. Simulation Levels:**

This module is linked to the Phobia Simulation module. For each fear, users will be provided with three simulations, namely easy, medium and hard level, to represent increasing fear intensity.

- 3. Professional Guidance**

This module allows users to request and/or be assigned counseling from therapists/psychiatrists for their fears. Therapists can view users' questionnaire results, and can communicate with assigned users using text messaging, and video chat.

- 4. Intensity Confirmation**

This module allows users to take a questionnaire to confirm the presence of their fear, and to determine the intensity level of their fear.

- 5. Progress Tracking**

This module allows users to determine their progress for exposure therapy using questionnaires. This module is linked to the User Profile Management Module, as only users who have made their profiles can access this module.

- 6. Payment Module**

This module provides therapists/psychiatrists with the ability to set up their payment details, and it lets users that need counseling with their exposure therapy to pay for the sessions.

- 7. User Profile Management**

This module provides Login, and Sign up facilities along with profile page management to both users, and therapists.

8. Feedback

This module allows users to give and view feedback for the simulations available for exposure therapy , and rate the therapist/psychiatrist that is assigned to the user for exposure therapy.

Types of Users and their Characteristics

The following table elaborates the different user classes that will use FearLESS, and their description.

Type of User	Description
Acrophobic (Heights) Users	Acrophobic people are those that have been diagnosed with acrophobia (fear of heights), and will use FearLESS to get exposure therapy for fear of heights. Such users, with the help of the simulations, will expose themselves to increasing heights until they have either overcome their fear, or feel the symptoms of their phobia to a lesser extent. They can also use FearLESS to get online counseling from the application's counseling module.
Nyctophobic (Darkness) Users	Nyctophobic people are those that have been diagnosed with nyctophobia (fear of darkness), and will use FearLESS to get exposure therapy for fear of darkness. Such users, with the help of the simulations, will expose themselves to increasing darkness until they have either overcome their fear, or feel the symptoms of their phobia to a lesser extent. They can also use FearLESS to get online counseling from the application's counseling module.
Inquisitive Users	These are users that are curious about whether or not they have a fear of heights and/or darkness. Such users will rely on the results given to them by the application's questionnaires, and their reactions to the simulations before they confirm if they have the fear or not.
Counselors (Therapists/Psychiatrists)	Therapists, psychiatrists or counseling professionals that are providing exposure therapy to people can use FearLESS's simulations to provide their patients exposure therapy from the safety of their offices. They can also use the questionnaires to determine their patient's fear intensity towards heights and/or darkness.
VR Enthusiasts	Such users will use FearLESS only to attain the VR experience.

Project Requirements

The following section describes the functional, non-functional requirements and overall scope of FearLESS.

Functional Requirements

The functional requirements for FearLESS includes:

1. Phobia Selection

- The system shall allow the user to access the simulation for nyctophobia (fear of darkness).
- The system shall allow the user to access the simulation for acrophobia (fear of heights).
- The system shall allow the therapist to access the simulation for the nyctophobia, and acrophobia.

2. Simulation Levels

- The system shall allow the user to access the easy level for the simulation of acrophobia.
- The system shall allow the user to access the medium level for the simulation of acrophobia.
- The system shall allow the user to access the hard level for the simulation of acrophobia.
- The system shall allow the user to access the easy level for the simulation of nyctophobia.
- The system shall allow the user to access the medium level for the simulation of nyctophobia.
- The system shall allow the user to access the hard level for the simulation of nyctophobia.

3. Counseling

- The system shall allow the user to chat with the appointed therapist.
- The system shall allow the user to do a video call with the appointed therapist.
- The system shall assign the users to the therapist.
- The system shall allow the therapist to view the results of the client's intensity identification test.
- The system shall allow the therapist to allow/deny the client's access to the simulation of its phobia.
- The system shall allow the user to request for changing the therapist assigned to it.
- The system shall allow the user to request for the appointment of the therapist.

4. Intensity Confirmation

- The system shall allow the user to take a questionnaire before simulation.
- The system shall allow the user to view the result of the intensity confirmation questionnaire.

5. Progress Tracking

- The system shall allow the registered user to take a questionnaire after the simulation.
- The system shall update the fear intensity level of registered users according to the result of the progress questionnaire.
- The system shall change the simulation level on the basis of fear intensity.

6. Users Profile Management

- The system shall allow the user to create an account.
- The system shall allow the user to login to its account using its email/username and password.
- The system shall allow the user to login to its account using its google account.
- The system shall allow the user to view its profile.
- The system shall allow the user to update its profile.
- If the user forgets its password, then the system shall allow the user to reset its password.

7. Payment

- The system shall allow the therapist to set payment details.
- The system shall allow the user to set payment details.
- The system shall allow the user to pay using subscription.

8. Feedback

- The system shall allow the user to give feedback for the application.
- The system shall allow the user to provide feedback to change the therapist.
- The system shall allow the user to view the feedback of the therapist.
- The system shall allow the user to view the feedback for the application.

Non-Functional Requirements

The non functional requirements for FearLESS are as follows:

1. Organizational requirements

- The system shall be developed using Unity version 2023.2.7f1 LTS version.
- The system shall be developed in C#.
- The VR will be set up using XR Interaction Toolkit version 2.5.2 and XR Plugin Management version 4.4.0

2. Security requirements

- The system shall provide user authentication and authorization to ensure authorized access.
- The system shall provide mechanisms for secure password recovery and management.

3. Usability requirements

- The system shall provide an intuitive interface that is easy to navigate in the desktop application.
- The simulations shall provide the users with guidance during the simulations for efficient interactions and experience.

4. Portability Requirements

- The system shall support compatibility with an Oculus Rift Headset.

Use Case Diagram

Based on FearLESS's intended functionality, the target audience for our application are users seeking exposure therapy for their fears, and counselors. These two sets of users have the following use cases available to them:

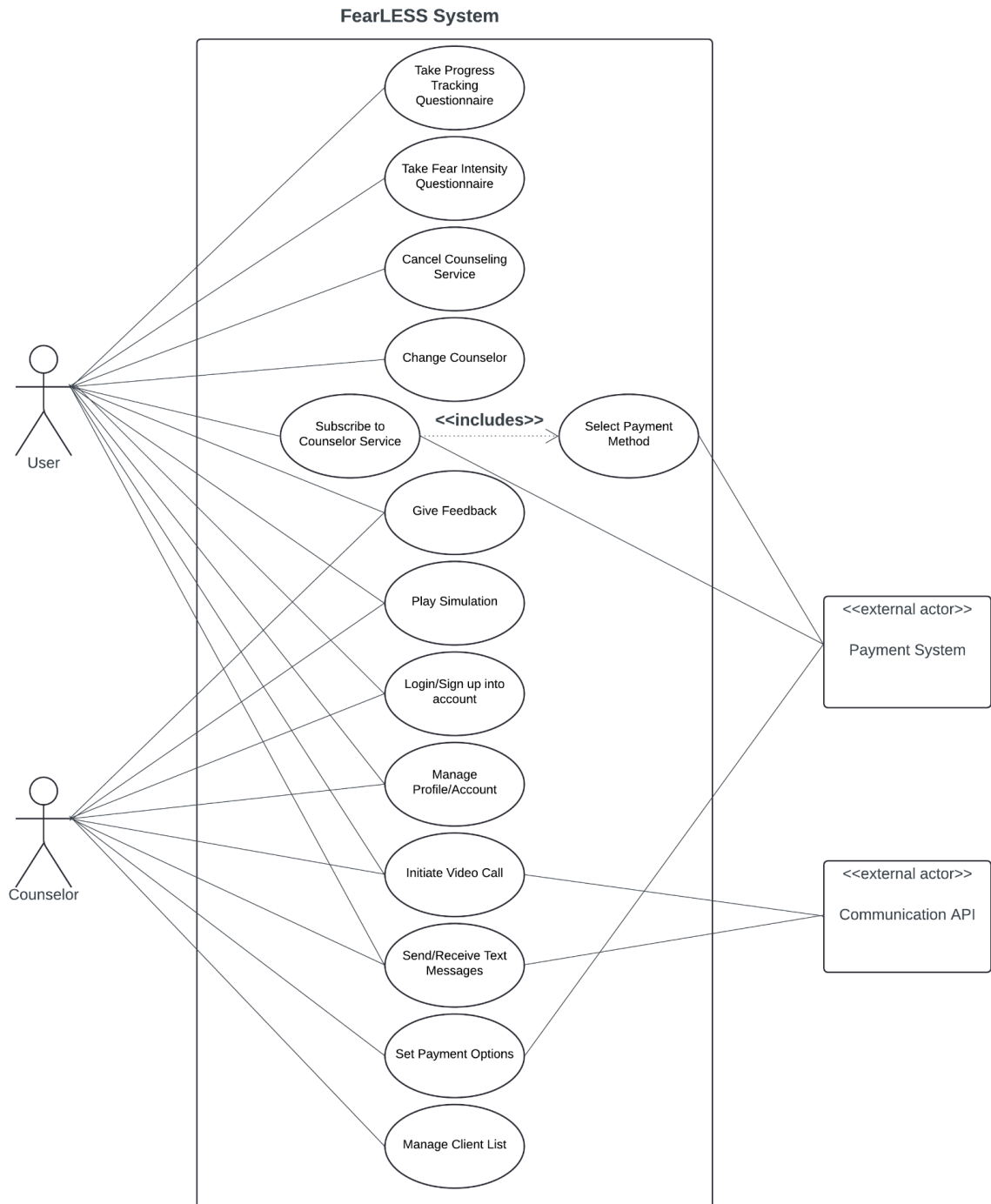


Figure 1: FearLESS Use Case Diagram

Detailed Use Cases

Take Progress Tracking Questionnaire

The following explains in detail the Take Progress Tracking Questionnaire use case

Use Case Name	UC1: Take Progress Tracking Questionnaire
Scope	FearLESS
Level	User-Level
Primary Actor	User
Stakeholders and Interests	- User: Wants to check if their fear intensity score improves after watching simulation
Pre-conditions	- FearLESS app is downloaded and running - User has a profile and is logged in - User has taken fear intensity questionnaire - User has watched a level of simulation - Questionnaires and their scores are present in the database
Success Guarantee	The user receives a score/result after taking the progress tracking questionnaire, and their fear score is updated
Main Success Scenario	1. User completes/exits the simulation level 2. User requests system to view the progress tracking questionnaire 3. User reads the question in the questionnaire 4. User selects an answer to the questionnaire 5. User moves on to the next question Repeat 3, 4 & 5 until all questions are answered 6. User submits their answers 7. User views their fear score after watching a simulation
Extensions	2a. User does not request to see the questionnaire 1. Fear score for that particular simulation session is marked as empty 3a. User does not answer the question in the questionnaire 1. The user reads the next question on the questionnaire 2. User selects answer or skips Repeat 1 and 2 until the end of the questionnaire 3. User is presented with all skipped questions 4. User is informed that they have to answer all questions in order to submit the form
Special Requirements	N/A
Technology and Data Variations List	N/A

Take Fear Intensity Questionnaire

The following explains in detail the Take Fear Intensity Questionnaire use case

Use Case Name	UC2: Take Fear Intensity Questionnaire
Scope	FearLESS
Level	User-Level
Primary Actor	User
Stakeholders and Interests	- User: Wants to check their fear intensity score for a particular fear
Pre-conditions	- FearLESS app is downloaded and running - User has selected a fear they want to take fear intensity for - Questionnaires and their scores are present in the database
Success Guarantee	The user receives a fear score
Main Success Scenario	1. User requests system to allow them to take the questionnaire for the fear 2. User reads the question in the questionnaire 3. User selects an answer to the questionnaire 4. User moves on to the next question Repeat 3, 4 & 5 until all questions are answered 5. User submits their answers 6. User views their fear score
Extensions	3a. User does not answer the question in the questionnaire 5. The user reads the next question on the questionnaire 6. User selects answer or skips Repeat 1 and 2 until the end of the questionnaire 7. User is presented with all skipped questions 8. User is informed that they have to answer all questions in order to submit the form
Special Requirements	N/A
Technology and Data Variations List	N/A

Subscribe to Counseling Service

The following explains in detail the Subscribe to Counseling Service use case

Use Case Name	UC3: Subscribe to Counseling Service
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Scope	FearLESS
Level	User-Level
Primary Actor	User, Payment API
Stakeholders and Interests	- User: They want to get guided counseling for their fear - Counselor: They want to assist user to overcome their fear of heights and/or darkness - Payment System: They want to ensure that a complete transaction is carried out between user and counselor
Pre-conditions	- FearLESS app is downloaded and running - User has a profile, and is logged in - User completed UC2: Take Fear Intensity Questionnaire - System is connected to Payment API
Success Guarantee	User is assigned to a counselor, and the user receives a proof of transaction.
Main Success Scenario	1. User requests system to assign them to a counselor 2. User views the counselor profiles and their payment options 3. User selects a counselor and clicks subscribe 4. The user selects a payment plan 5. <<includes>> Select Payment Method (UC4) 6. Payment system processes the payment 7. The subscription is confirmed and the user continues with their sessions.
Extensions	2a. Counselor is not available/has no more space 1. User is asked to select a new counselor 5a. The payment method is declined 1. User is prompted to select and enter a different payment method 5b. The user aborts the transaction 1. No payment is made
Special Requirements	- The system must support payment gateways like Stripe, and must be compatible with these third party APIs and they must be secure
Technology and Data Variations List	- User must be presented with the different payment gateways for subscription

Select Payment Method

The following explains in detail the Select Payment Method use case

Use Case Name	UC4: Select Payment Method
Scope	FearLESS
Level	User-Level
Primary Actor	User, Payment API
Stakeholders and Interests	- User: They want to get guided counseling for their fear - Counselor: They want to assist user to overcome their fear of heights and/or darkness - Payment System: They want to ensure that a complete transaction is carried out between user and counselor
Pre-conditions	- FearLESS app is downloaded and running - User has a profile, and is logged in - User completed UC2: Take Fear Intensity Questionnaire - User has selected a counselor - User has selected a payment plan - System is connected to the Payment API
Success Guarantee	User selects a payment method and the system validates their payment
Main Success Scenario	1. The user is presented with payment options for counseling 2. User selects a payment method option 3. User inputs their payment details based on the option selected 4. Payment process continues
Extensions	3a. Payment information is invalid 1. The user is informed that their payment information is invalid 2. The user is asked to reenter the payment details
Special Requirements	- The system must ensure that the payment process is secure - The system must support payment gateways like Stripe, and must be compatible with these third party APIs
Technology and Data Variations List	- User must be presented with the different payment gateways for subscription

Change Counselor

The following explains in detail the Change Counselor use case

Use Case Name	UC5: Change Counselor
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to start session with a new counselor - Counselor: Wants to assist user to overcome their fear of heights and/or darkness
Pre-conditions	- FearLESS app is downloaded and running - User has a profile, and is logged in - User is already assigned to a counselor
Success Guarantee	User is assigned to a new counselor, and is removed from the list of clients from the previous counselor's list.
Main Success Scenario	1. User requests system to change counselor 2. System confirms with user about this decision 3. System presents user with the counselor list 4. System removes user from the previous counselor's services 5. Counselor is informed that a user is removed from their list. 6. User selects a new counselor 7. User subscribes with the new counselor (UC3 and UC4)
Extensions	2a. User does not confirm 1. User is still subscribed to the previous counselor 4a. System is unable to remove user from the counselor's list 1. User is informed and is asked to repeat the process 5a. Counselor is not available/has no more space 2. User is asked to select a new counselor
Special Requirements	N/A
Technology and Data Variations List	N/A

Cancel Counseling Service

The following explains in detail the Cancel Counseling Service use case

Use Case Name	UC6: Cancel Counseling Service
Scope	FearLESS
Level	User-Level

Primary Actor	User, Counselor
Stakeholders and Interests	- User: No longer wants the counselor's services - Counselor: Wants to be informed about no longer servicing a particular user
Pre-conditions	- FearLESS app is downloaded and running - User has a profile and is logged in - User is currently subscribed to a therapist
Success Guarantee	User is removed from the counselor's list
Main Success Scenario	1. User requests the system to cancel counseling 2. User confirms decision to cancel counseling 3. User is removed from the counselor list of clients 4. User is informed of removal
Extensions	2a. User does not confirm cancellation 1. User is still subscribed to the counseling 3a. User is not removed from the counselor's client list 1. User is asked by the system to initiate cancellation again
Special Requirements	N/A
Technology and Data Variations List	N/A

Give Feedback

The following explains in detail the Give Feedback use case

Use Case Name	UC7: Give Feedback
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to give feedback on simulations, app and/or counselor - Counselor: Wants to give suggestions about the simulations, and/or application
Pre-conditions	- FearLESS application is downloaded and running
Success Guarantee	User/Counselor feedback is submitted
Main Success Scenario	1. User/Counselor request system to give feedback 2. User is asked what they want to give feedback on

	(application, simulation or counselor) 3. User/Counselor selects their feedback option 4. User writes down their suggestions and/or opinions 5. User submits feedback after review 6. User is informed about their feedback submission
Extensions	2a. The name of simulation level/counselor name is not updated in the feedback list 1. User is asked to manually write the name of the simulation or counselor they are feedbacking on
Special Requirements	N/A
Technology and Data Variations List	N/A

Play Simulation

The following explains in detail the Play Simulation use case

Use Case Name	UC8: Play Simulation
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to watch the simulations to get exposure therapy or explore the VR experience - Counselor: Wants to show their clients simulations for exposure therapy
Pre-conditions	- FearLESS application is downloaded and running - If users, then they have taken a fear intensity questionnaire. - Users have a headset that is plugged into their system - System contains the relevant simulations and their levels
Success Guarantee	Users watch the simulations
Main Success Scenario	1. System presents the user with simulations based on the user's fear score from UC2 2. User selects a simulation level that they are allowed to watch (counselors can watch all so they select one) 3. User puts on their headset and watches the simulation
Extensions	1a. Due to their fear score, none of the simulation levels are available to the user

	1. User is requested to get counseling from UC3 2. Counselor conducts sessions with user 3. Once counselor approves, the user watches the simulations
Special Requirements	- Users must have a system that supports VR simulations - VR headset must be connected to the system before watching the simulations
Technology and Data Variations List	- User's VR equipment must include a headset (with microphones preferably), and controllers

Login/SignUp Into Account

The following explains in detail the Login/SignUp into Account use case

Use Case Name	UC9.1: Login to Account
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to Login so that they can access their profile, allow progress tracking, and get counseling - Counselor: Wants to Login so that they can access profile, and guide their clients to overcome their fears
Pre-conditions	- FearLESS is downloaded and running - The user/counselor has already signed in (UC 9.2)
Success Guarantee	The user/counselor successfully logs in and can access their account
Main Success Scenario	1. User requests system to log in 2. User inputs their email 3. User inputs their password 4. User submits credentials and system confirms 5. User is now logged in
Extensions	4a. User has submitted the wrong credentials 1. User is asked to log in again 2. If user has forgotten password, they inform the system a. System sends user an email with option to set new password 3. User inputs email, and password again
Special Requirements	N/A

Technology and Data Variations List	N/A
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Use Case Name	UC9.2: Sign into Account
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to Sign in so that they can access their profile, allow progress tracking, and get counseling - Counselor: Wants to Sign in so that they can access profile, and guide their clients to overcome their fears
Pre-conditions	- FearLESS is downloaded and running
Success Guarantee	The user/counselor successfully signs in and can access their account
Main Success Scenario	1. User/Counselor requests system to create an account 2. User/Counselor inputs their email 3. User/Counselor inputs their password 4. User/Counselor re-enters password for reconfirmation 5. If Counselor, they enter their medical identification ID, and maximum amount of clients they want to service (if they have a cap) 6. User/Counselor submits credentials and system confirms 7. User/Counselor now has an account/profile with FearLESS
Extensions	6a. User/Counselor has submitted the wrong credentials 1. User is asked to log in again 2. If user has forgotten password, they inform the system a. System sends user an email with option to set new password 3. User inputs email, and password again
Special Requirements	N/A
Technology and Data Variations List	N/A

Manage Profile Account

The following explains in detail the Manage Profile Account use case

Use Case Name	UC10: Manage Profile Account
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Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor
Stakeholders and Interests	- User: Wants to view, and update their profile information - Counselor: Wants to view and update their profile information
Pre-conditions	- FearLESS application is downloaded and running - User/Counselor already has an profile
Success Guarantee	User/Counselor have successfully updates/viewed their profile details
Main Success Scenario	1. The user/counselor logs into their profile using email, and password 2. User/Counselor requests system to view their profile 3. User/Counselor requests system to update their password 4. User/Counselor inputs new password 5. The system informs the user/counselor that their password is updated
Extensions	1a. The user cannot log in 1. System asks the user/counselor to input their email and password again.
Special Requirements	N/A
Technology and Data Variations List	N/A

Initiate Video Call

The following explains in detail the Initiate Video Call use case

Use Case Name	UC11: Initiate Video Call
Scope	FearLESS
Level	User-Level
Primary Actor	User, Counselor, Communication API
Stakeholders and Interests	- User/Counselor: Want to start a video call session with each other so that they can communicate with each other about the user's fears and how to overcome

	them
Pre-conditions	- FearLESS app is downloaded and running - User/Counselor have a profile and are logged in - User has subscribed to the counselor's service - System is connected to Communication API
Success Guarantee	User and Counselor conclude their video call with each other successfully
Main Success Scenario	1. User/Counselor informs the other about starting a video call 2. User/Counselor requests the communication API to start the video call 3. User and Counselor have their session via video call 4. When session is completed, user and counselor exit the call
Extensions	2a. Network Issues prevent user/counselor from starting the call: 1. System informs the user/counselor of the network issue with the message "Poor network. Try Again" 2b. User/Counselor are unavailable: 1. Counselor/User can wait for the other to join the call 2. Counselor/User can exit call whenever they please 2c. User/Counselor declines the call: 1. System informs user/counselor of call decline, and ends the call 2d. API failure 1. System informs user about the issue with the communication API
Special Requirements	- System should ensure compatibility with API - There should be an audio only option if video call fails
Technology and Data Variations List	- Different Wifi bandwidth can affect video call quality - The PC used by the user/counselor should be able to support video or at least an audio call.

Send/Receive Text Messages

The following explains in detail the Send and Receive Text Messages use case

Use Case Name	UC12: Send/Receive Text Messages
Scope	FearLESS
Level	User-Level

Primary Actor	User, Counselor, Communication API
Stakeholders and Interests	- User/Counselor: Want to start a text message conversation so that they can communicate with each other about the user's fears and how to overcome them
Pre-conditions	- FearLESS app is downloaded and running - User/Counselor have a profile and are logged in - User has subscribed to the counselor's service - System is connected to communication API
Success Guarantee	User and Counselor can send and receive text messages from each other successfully
Main Success Scenario	1. User/Counselor requests the system to open the chat with the other user. In case of counselor, they select the user they want to communicate with 2. User/Counselor sends a text to the other 3. Counselor/User reply to the text Repeat steps 2 and 3 until User and Counselor have completed their communication
Extensions	2a. Message delivery to the other user fails due to network or API issues 1. The system informs the user of the failure 2. The system retries to send the message 2b. Message takes too long to deliver 1. System informs user of late delivery 2. System shows user loading message to show that message is being sent
Special Requirements	- System should ensure compatibility with API
Technology and Data Variations List	- Different Wifi bandwidth can affect delivery time

Set Payment Options

The following explains in detail the Set Payment Options use case

Use Case Name	UC13: Set Payment Options
Scope	FearLESS
Level	User-Level
Primary Actor	Counselor, Payment API
Stakeholders and Interests	- Counselor: Wants to set the price rate for each session

	or a group of sessions. - Payment API: Wants to ensure that the payment options set by the counselor are valid, processed and stored
Pre-conditions	- FearLESS app is downloaded and running - Counselor has a profile and is logged in - System is connected to the payment API
Success Guarantee	The counselor's payment options are successfully set and the users can view their payment options as well
Main Success Scenario	1. Counselor requests the system to view their profile page 2. Counselor views profile page and selects "Set Payment Option" button 3. System displays relevant field to fill (for e.g. available dates, price rate per session/week/month, payment method e.t.c) 4. Counselor fills the data for each section especially the price rate and method 5. System verifies the payment option details entered by the counselor 6. Counselor verifies detail 7. System saves payment options and views them on the counselor's profile page Repeat set 2, 3, 4, 5, 6 and 7 to update any payment option information
Extensions	4a. Counselor fills in invalid (over the limit) rates or misses a field of input 1. System informs counselor of the error using an error message 5a. Payment API does not support/discontinued a particular payment method 1. System informs counselor of the change, and reminds them to update payment method information 7a. System fails to save the counselor's payment option information 1. System sends counselor an error message
Special Requirements	- Payment API must be compatible with the system
Technology and Data Variations List	N/A

Manage Client List

The following explains in detail the Manage Client List use case

Use Case Name	UC14: Manage Client List
Scope	FearLESS
Level	User-Level
Primary Actor	Counselor
Stakeholders and Interests	- Counselor: Wants to know which users have requested/ended services with them, and they want to keep track of their current users. They may also want to control how many users they want to help
Pre-conditions	- FearLESS app is downloaded and running - Counselor has a profile and is logged in
Success Guarantee	Counselor can view, and update (add/remove) their client list
Main Success Scenario	1. Counselor requests the system to view the client list 2. System displays the client list to the counselor 3. Counselor requests system to view the waitlist/request list for users asking for their service. 4. Counselor requests system to view a user's profile (containing fear score) 5. Counselor requests system to add user to their list 6. System confirms and updates client list
Extensions	2a. Particular user subscribes to the counselor have left their service 1. System notifies the counselor that the client has left their service 2. If they want, counselor can ask system to remove that user's chats and video logs with the user 5a. Counselor has reached their max number of clients 1. System notifies the client that they have reached max number of clients they can service 2. Counselor returns to client list or requests system to add the client to their list anyways 3. System updates the counselor's capacity, and client list 6a. System fails to update the client list 1. System notifies the counselor of the error
Special Requirements	N/A
Technology and Data Variations List	N/A

Domain Model

The following domain model represents how external entities, such as the player, and counselor, interact with each other, and with internal entities such as the simulations and questionnaires, within the context of the FearLESS application.

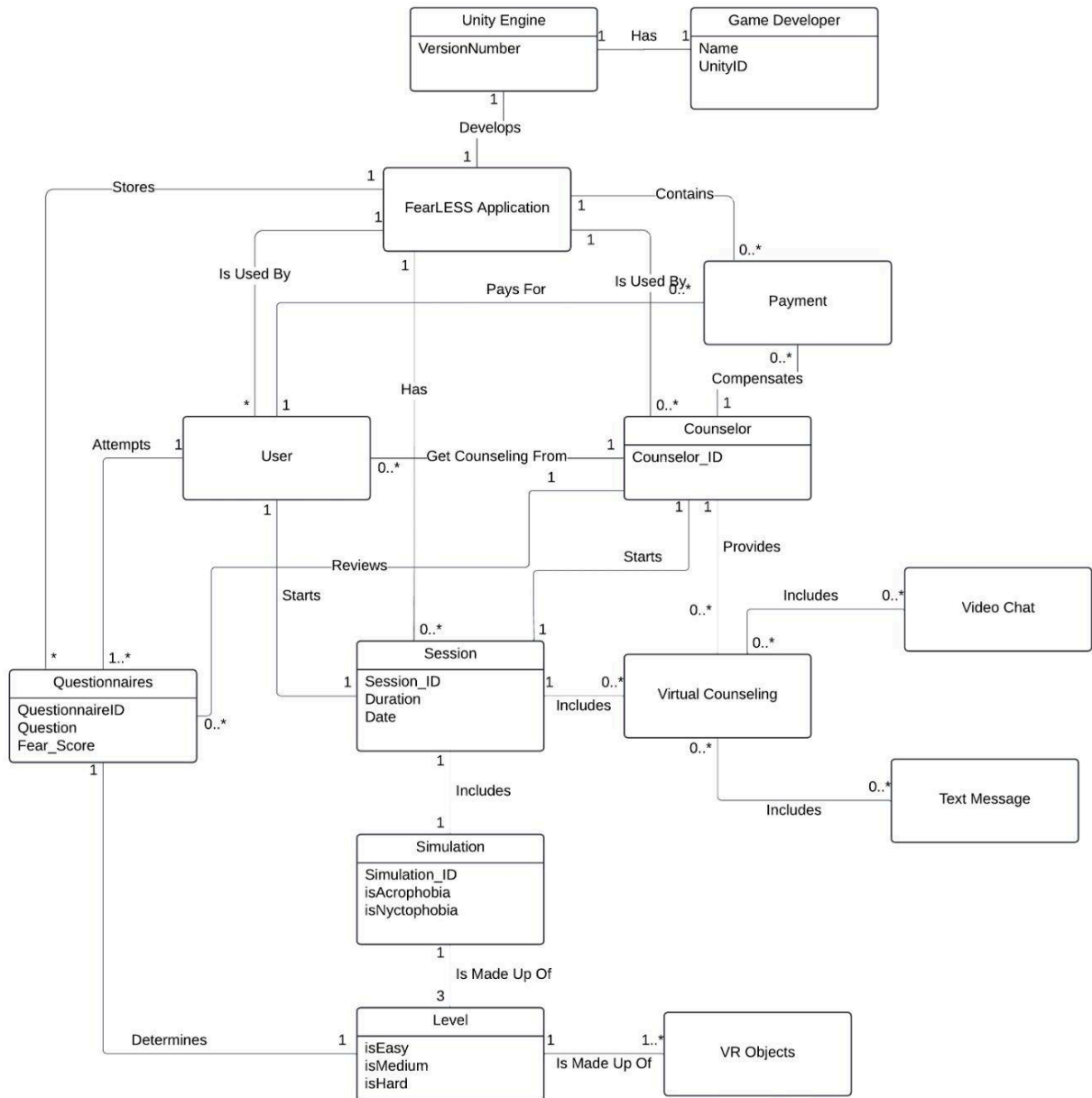


Figure 2: FearLESS Domain Model

System Overview

The following section contains design documentation and diagrams that best illustrate FearLESS's functionality. The architectural design diagram, and data flows represents the architecture, and data sinks/sources of the entire system respectively. Meanwhile the design models including activity diagrams, system level sequence diagrams, and state transition diagrams represent the profile management module of the FearLESS application, as it was part of the first iteration deliverables.

Systems Architectural Design

The architecture pattern that best describes our application FearLESS is microservices, as shown in the diagram below, and this is for several reasons. Firstly, FearLESS is composed of several modules/features, such as the therapist services, payment and simulation services, that can be developed and implemented independently. Secondly, microservices gives us an opportunity to add new services and scale existing services without affecting the current functionality and integrity of the application. For example, having a VR service, in the Player Session Microservice, gives us the flexibility to add more levels or simulations for other phobias into our application. Finally, FearLESS is also integrated with third party APIs such as Twilio (for real-time video chatting and text messaging), and PlayFab (for data storage), which can be easily integrated, due to the microservice architecture.

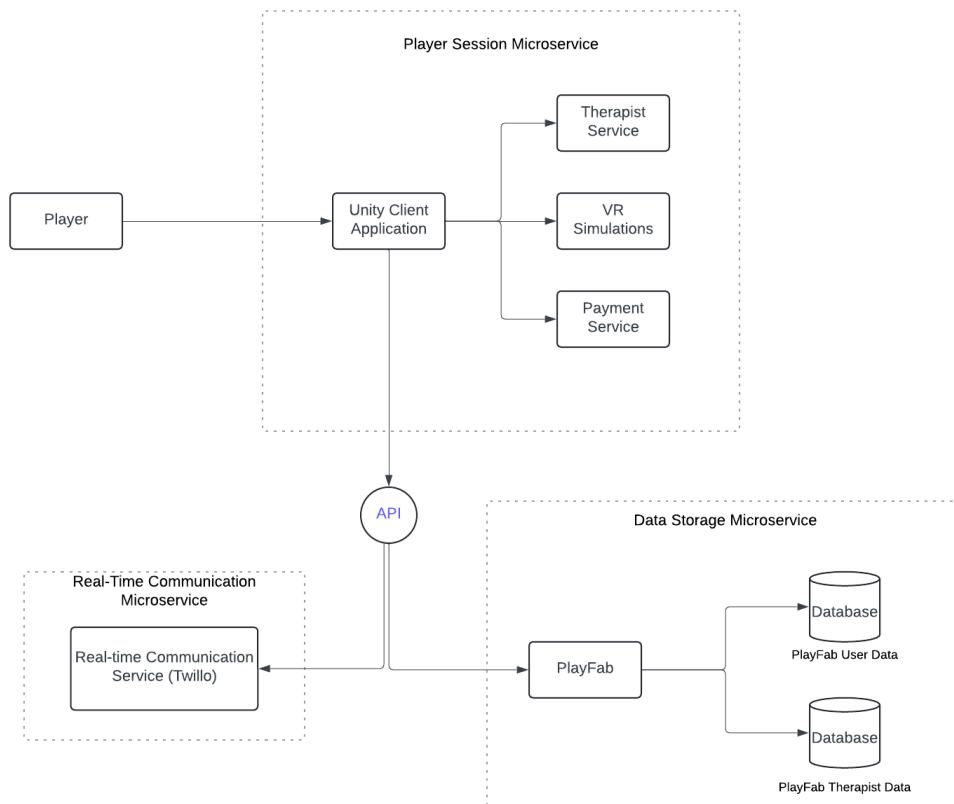


Figure 3: System Architecture Diagram

Design Models

The following design diagrams vital for the system, and profile management module's understanding are as follows:

Activity Diagram

The following are activity diagram for the profile management module:

Signup/Registration Activity Diagram

The following is activity diagram for sign up into FearLESS

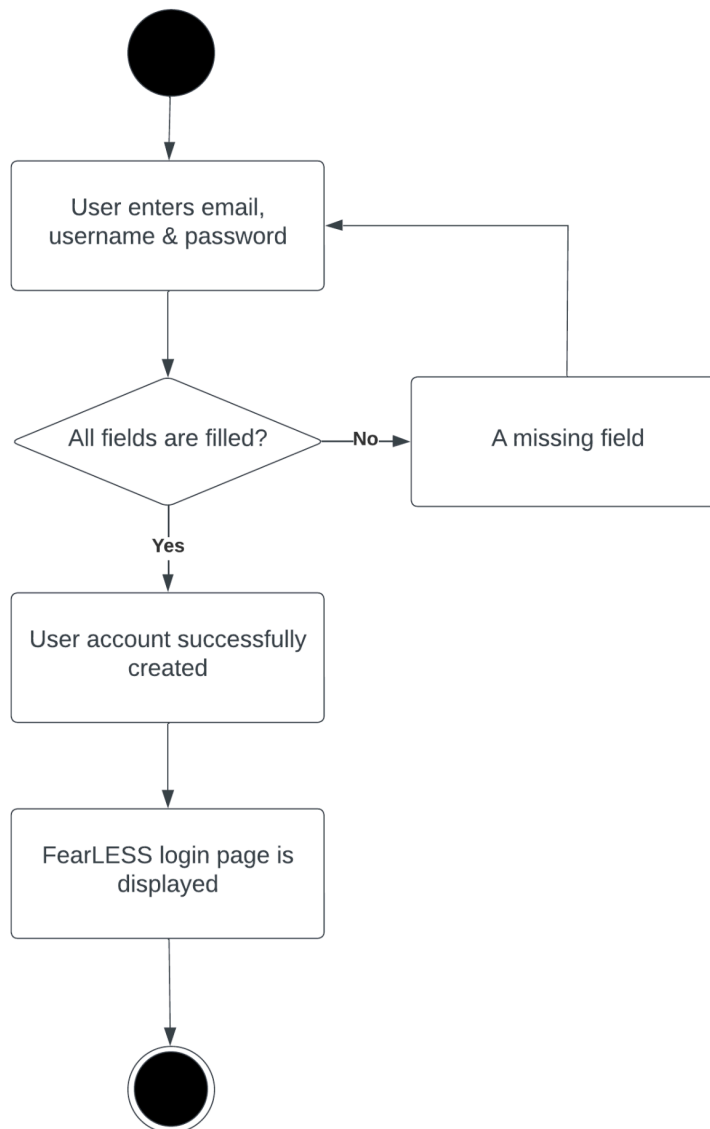


Figure 4: Activity Diagram for Sign Up

Login Activity Diagram

The following is activity diagram for logging into a user's FearLESS profile

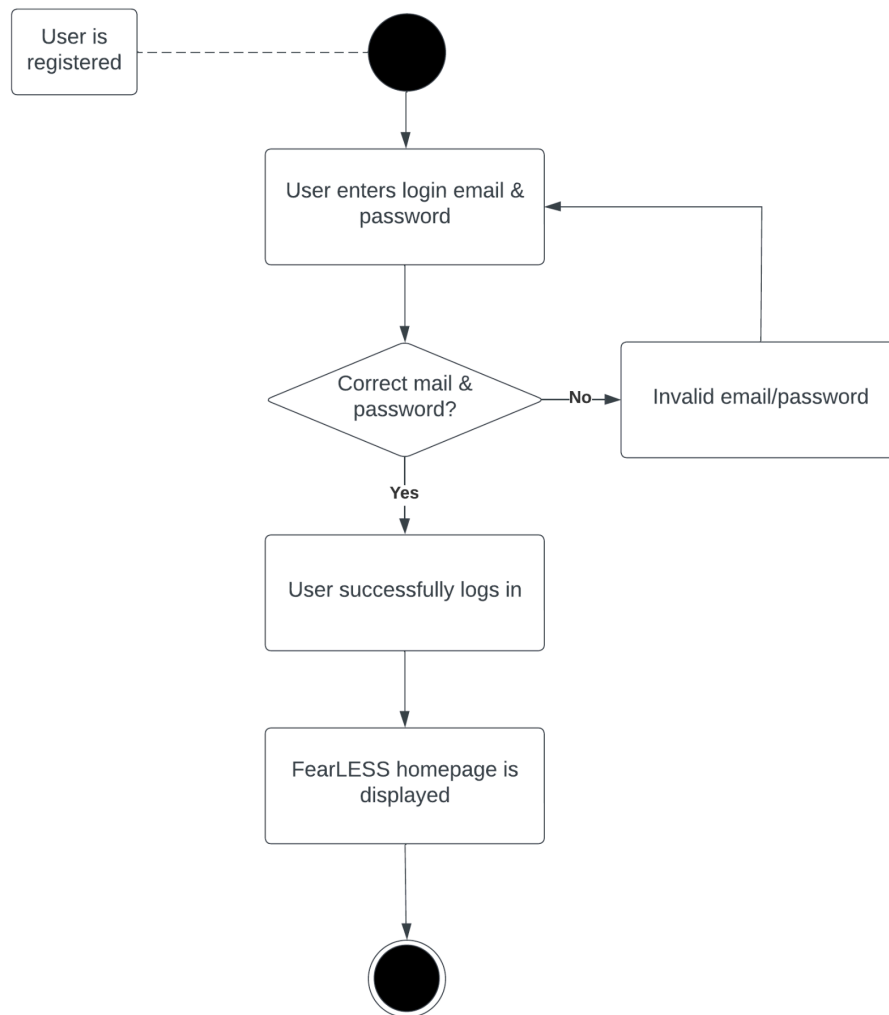


Figure 5: Activity Diagram for Login

Edit Profile Activity Diagram

The following is activity diagram for profiling management, specifically editing profile details

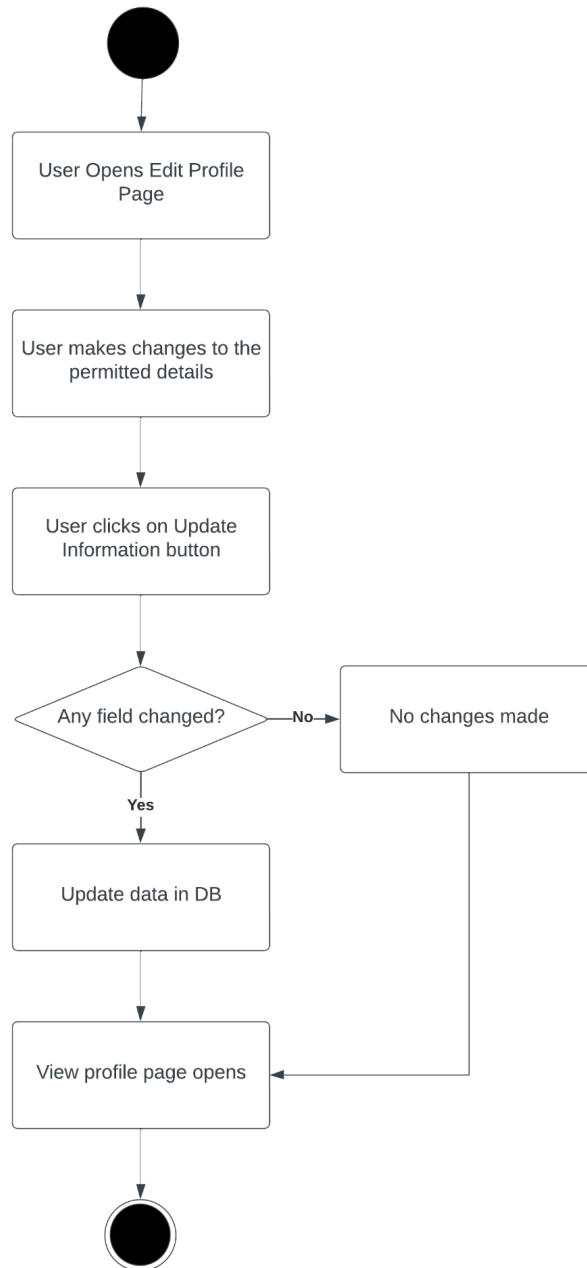


Figure 6: Activity Diagram for Edit Profile

Data Flow Diagram

Data Flow diagrams provide a comprehensive, understandable and dynamic view of the system. It shows the processing of data in the system. We made the DFDs up to level 1.

Below is the level 0 data flow diagram representing the high level flow of data without diving into details of processes of the system.

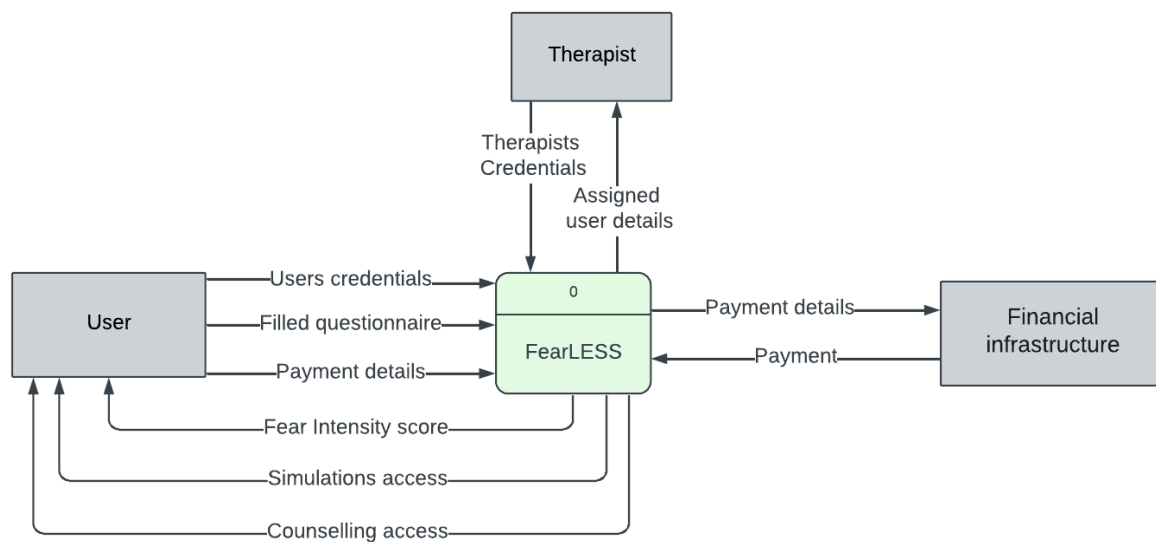


Figure 7: Data Flow Diagram (Level 0)

Below is the level 1 data flow diagram providing a detailed view of the processes and data stores of the entire system.

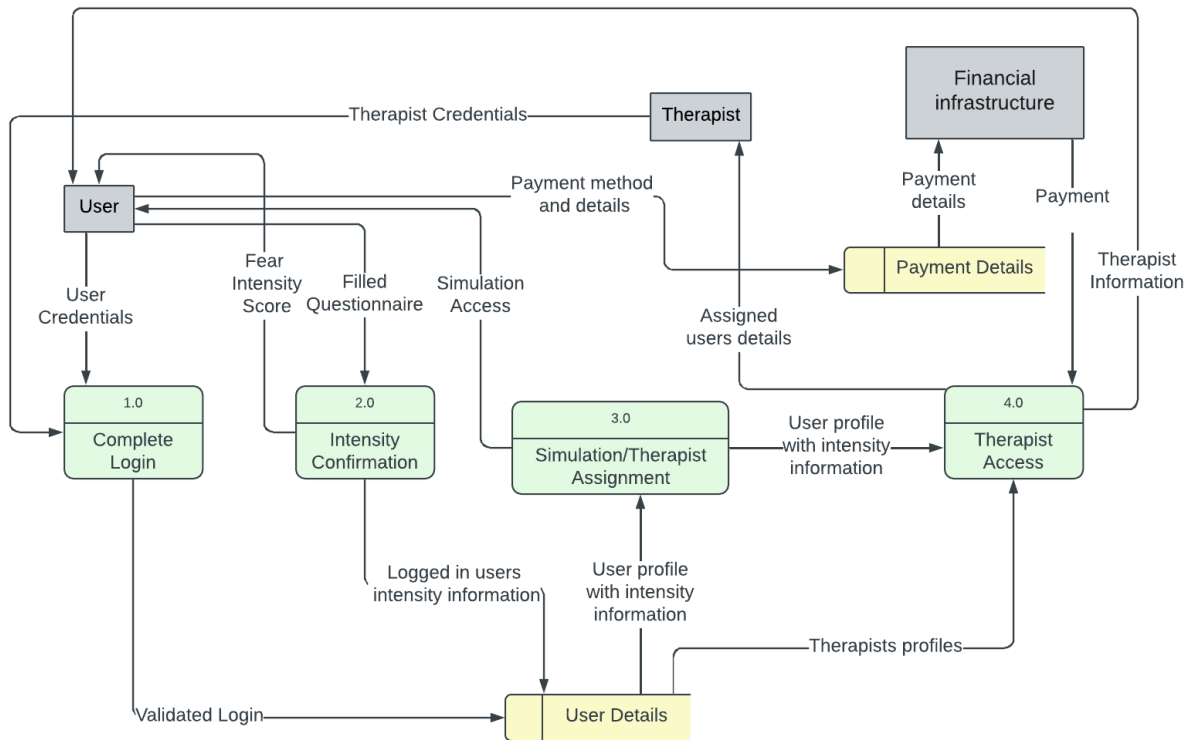


Figure 8: Data Flow Diagram (Level 1)

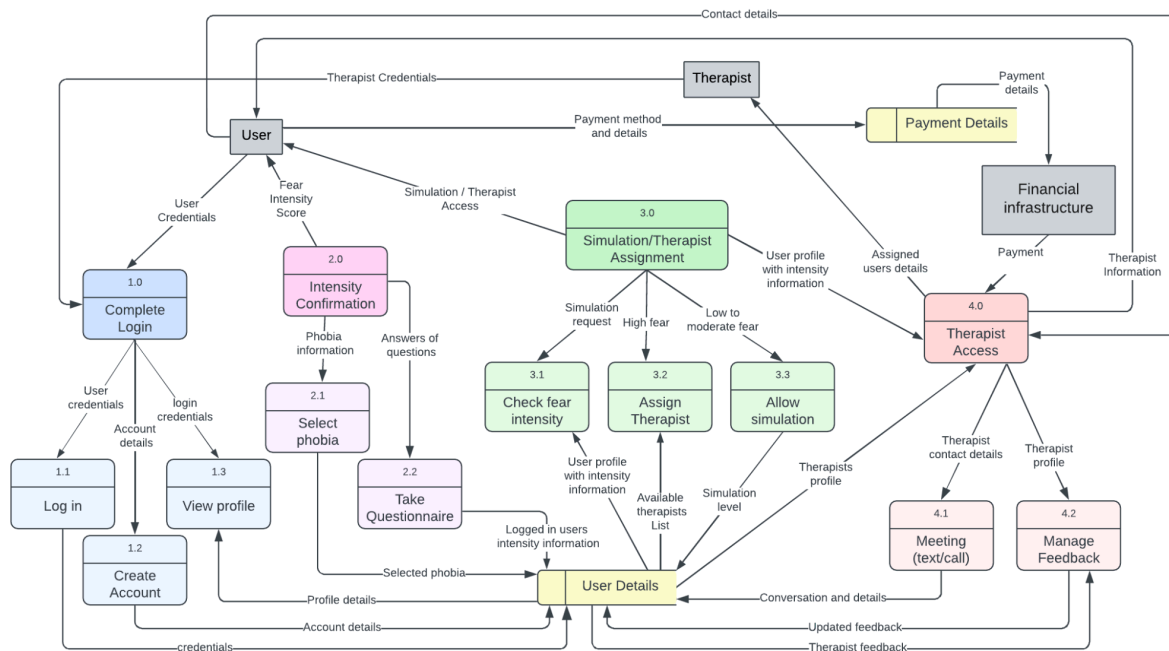


Figure 9: Data Flow Diagram (Level 2)

System Level Sequence Diagram

The system sequence diagram represents a clear view of how the user and system interacts with each other. Below is the system sequence diagram showing interaction of the user with the system while login, signup and update account.

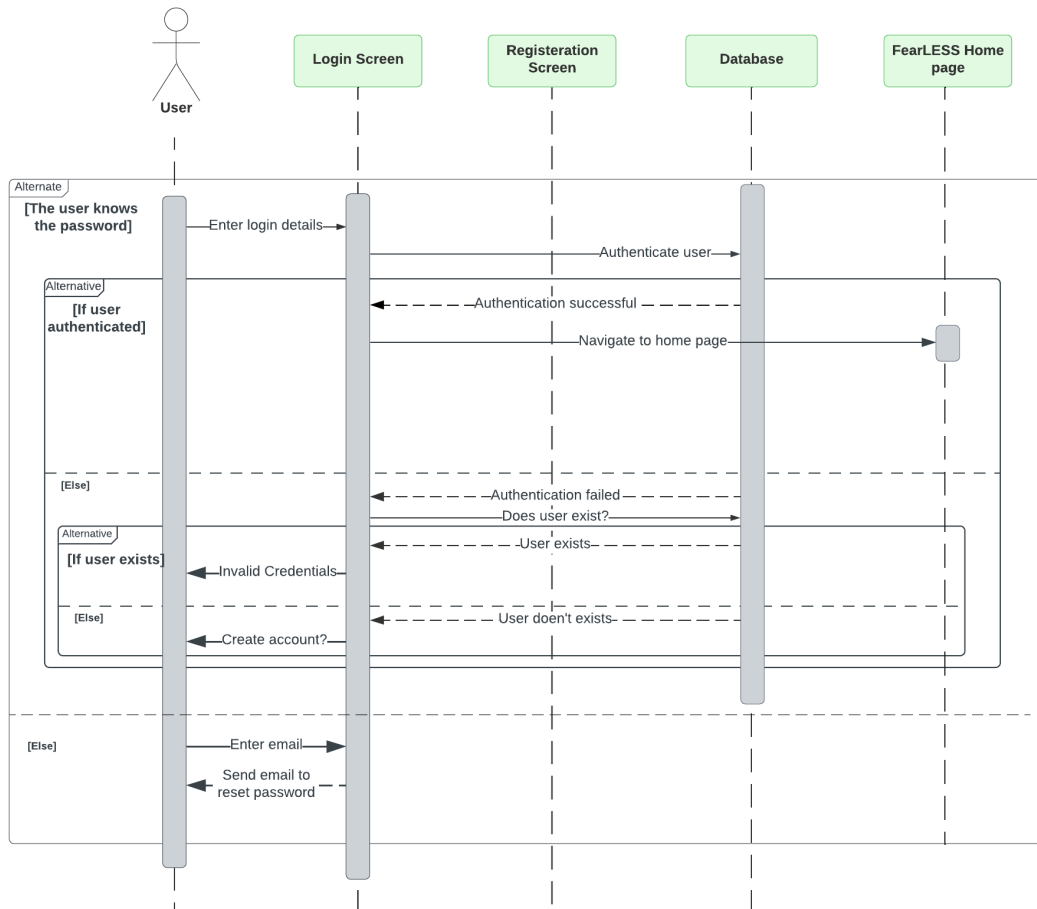


Figure 10: System Level Sequence Diagram for Login

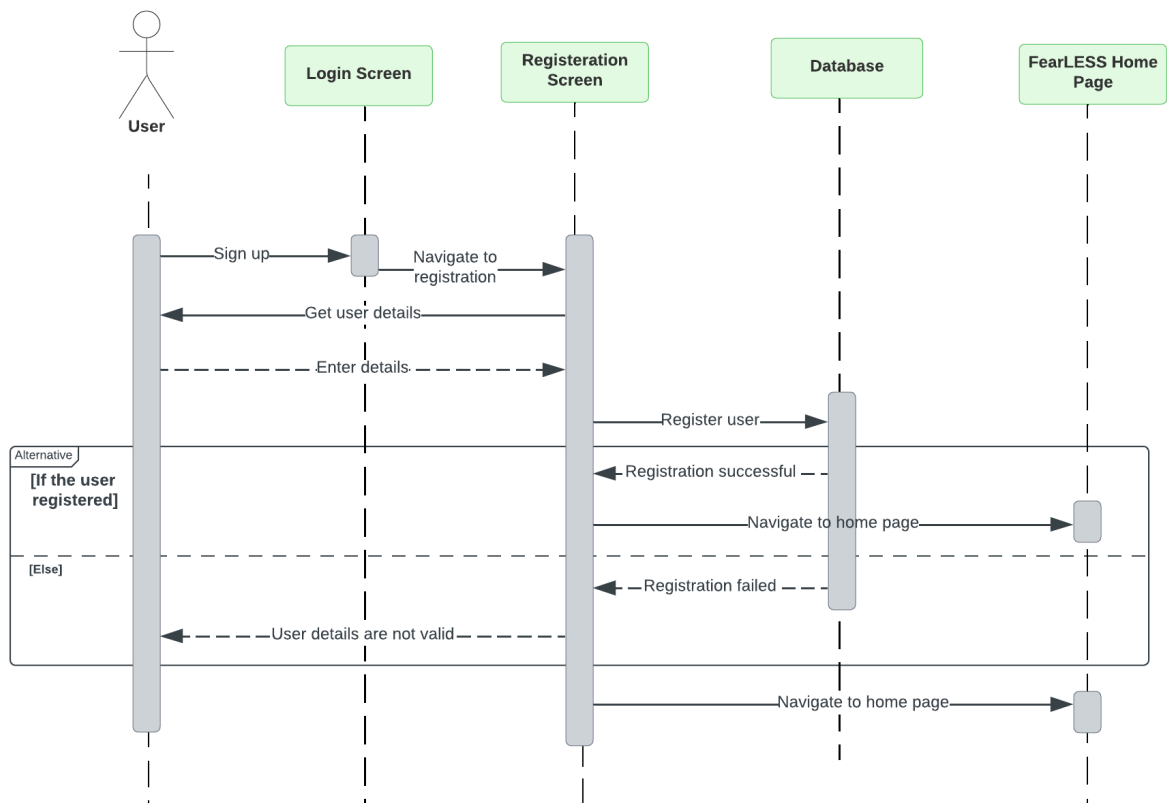


Figure 11: System Level Sequence Diagram for Signup

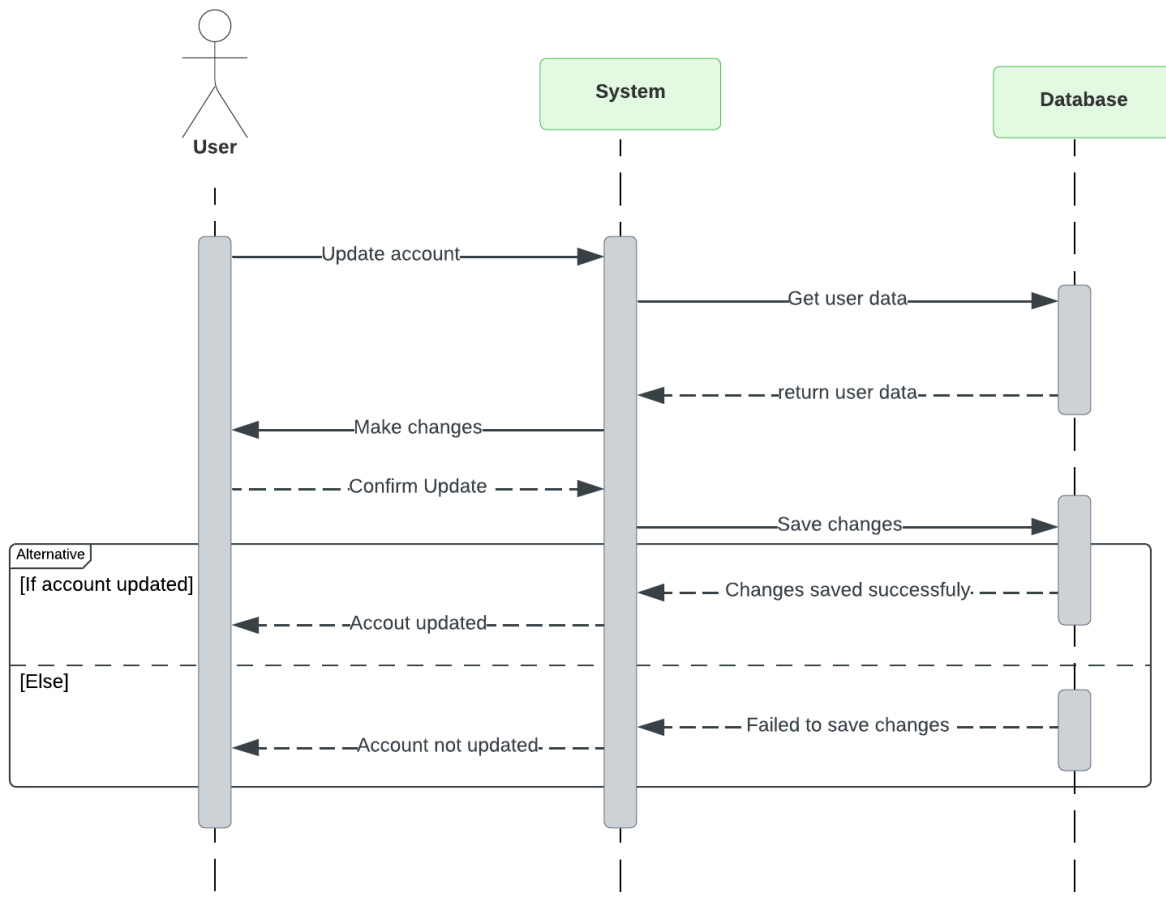


Figure 12: System Level Sequence Diagram for Update Account

State Transition Diagram

Below is the state transition diagram representing the states in user management and the transitions between the states.

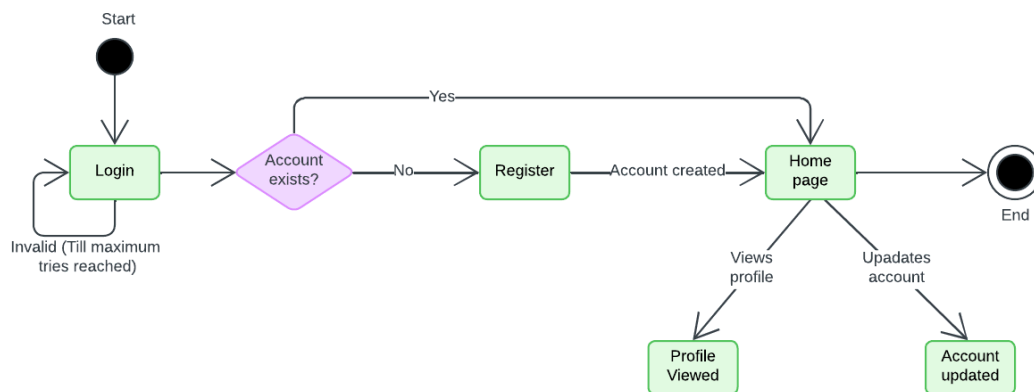


Figure 13: State Transition Diagram for User Management

The following EERD (Extended Entity Relationship Diagram) represents how the different entities, in FearLESS, are related to and interact with each other, and it also shows the form in which the entity's data and the relationship's data is stored. Notably, the ERD clearly identifies how a player receives exposure therapy for their fears with help of sessions in the form of simulations or in the form of video/text messaging with a therapist the player is subscribed to. The ERD also indicates that no credit card/payment details are stored since such information is sensitive, and storing it would pose a security issue. Furthermore, when a player attempts a questionnaire their fear score is calculated on the basis of their answers in the questionnaires. Note that the "TestPurpose" attribute attached to the "Attempts" relationship is indicative of the system's way of identifying whether an attempt was for fear intensity confirmation or for progress tracking.



First Iteration Implementation

According to our timeline, our first iteration includes the implementation of user profile management, and of the entire desktop application using Figma both of which have been implemented. Additionally, in preparation of iteration 2 we have collected certain assets for the Acrophobia simulations along with fear intensity confirmation questionnaires for acrophobia which have been approved by psychiatrists. The Game Design Document detailing simulations for all levels for Acrophobia and Nyctophobia was also completed in this iteration.

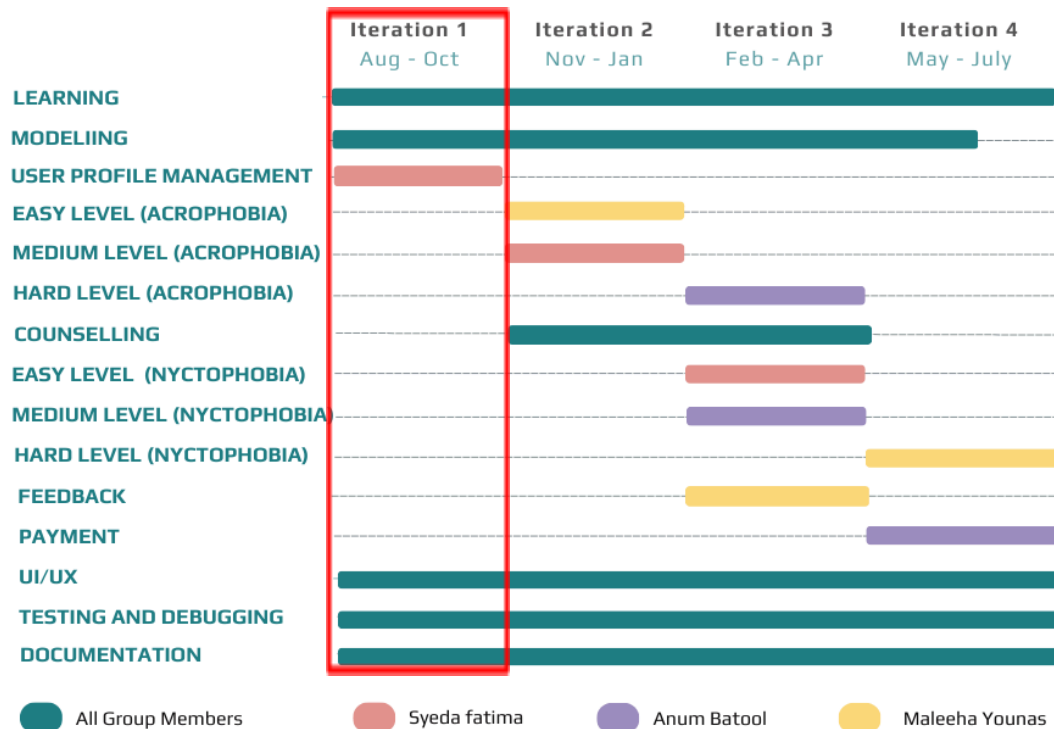


Figure 15: FearLESS Development Timeline

Profile Management Implementation

This section is dedicated to the profile management panels that have been implemented to completion in Unity. This includes signup/creation of profile, login, forgot password functionality during login, and view/update profile.

Login Panel

We have implemented the login page that takes the user's email and password as input. The user is informed when the input has given the wrong credentials. Clicking "Login" leads to the main page.

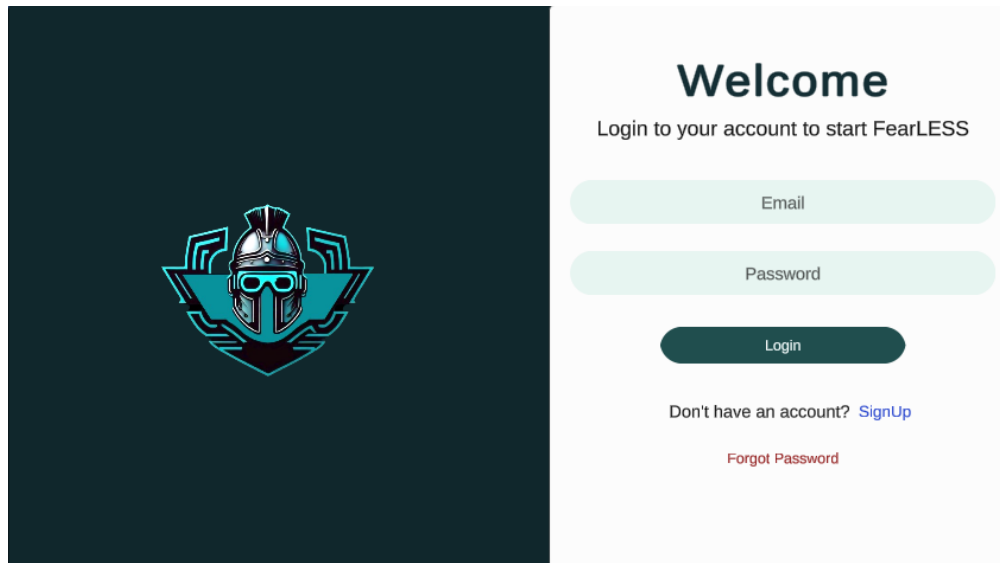
The login page features a dark blue sidebar on the left with a teal and black logo of a helmet with goggles. The main content area is light gray and contains the heading "Welcome" in bold black text, followed by the subtext "Login to your account to start FearLESS". Below this are two light green input fields labeled "Email" and "Password". A dark green "Login" button is positioned below the fields. At the bottom, there is a link "Don't have an account? [SignUp](#)" in blue and a link "Forgot Password" in red.

Figure 16: Login Page

Sign Up/Registration Panel

We have implemented the sign up page that takes the user's email and password as input, and makes the user re-confirm the password. Clicking "Signup" leads to the main page.

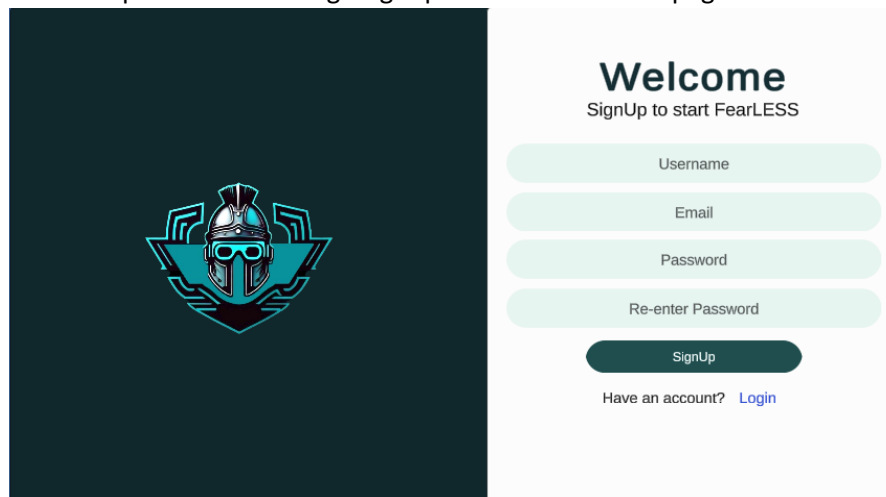
The signup page features the same dark blue sidebar with the logo as the login page. The main content area is light gray and contains the heading "Welcome" in bold black text, followed by the subtext "SignUp to start FearLESS". Below this are four light green input fields labeled "Username", "Email", "Password", and "Re-enter Password". A dark green "SignUp" button is positioned below the fields. At the bottom, there is a link "Have an account? [Login](#)" in blue.

Figure 17: Signup/Registration Panel

Forget Password Panel

For the forget password functionality, after the user requests to change password, the user enters an email address which contains their recovery email address.

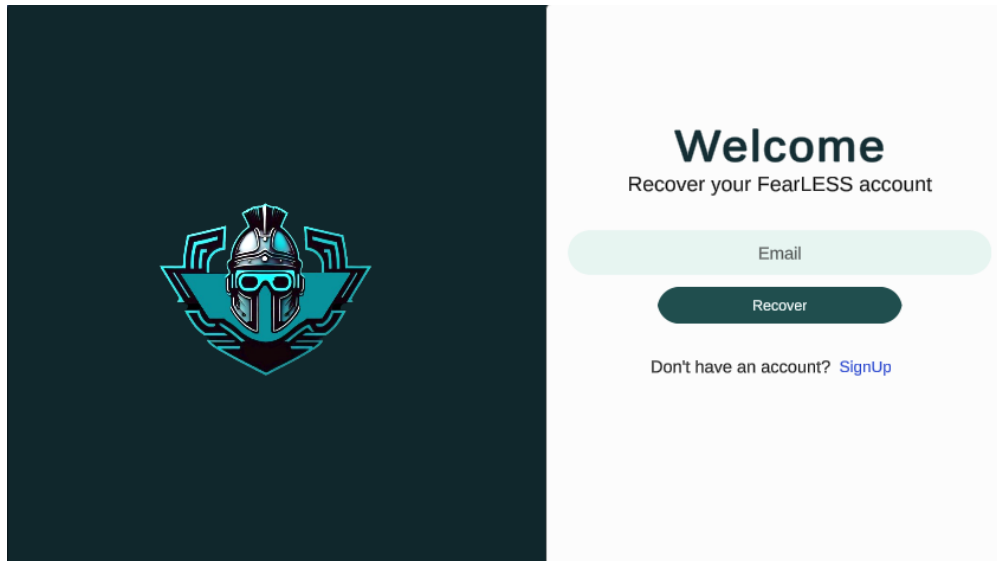


Figure 18: Forget Password Panel

View Profile Panel

The following image is the profile page that contains the user's information.

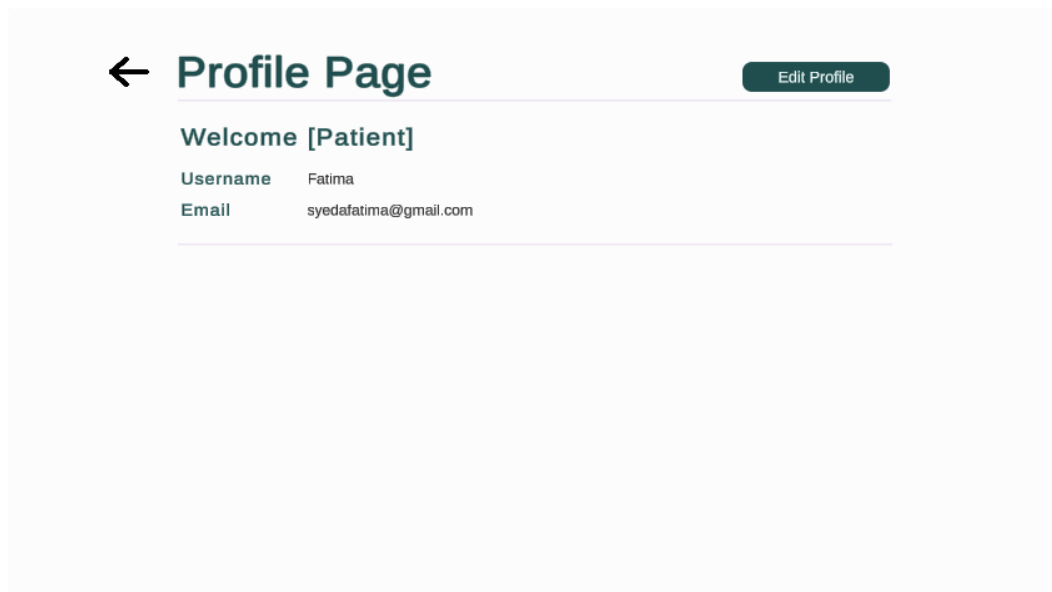


Figure 19: View Profile Panel

Edit Profile Panel

The following panel appears once the user clicks on the Update Profile Button. This allows the user to change their username, password and email address.

← Profile Page

Username	Fatima
Password	***** 
Email	syedafatima0998@gmail.com

Update Profile

Figure 20: Edit Profile Panel

Asset Collection

For the three levels of acrophobia we collected some online available assets in addition to self created assets.

Easy level

- **Mountains/hills:** In the easy level simulations, there will be mountains.
- **Flowers:** There will be flowers in the mountain meadow of easy level simulations.
- **Shed:** In easy level simulations there will be a wooden shed.
- **Wooden fence:** In easy level simulations there will be a wooden fence on the edge of the crest of the mountain.
- **Trees:** The trees in the mountain meadows.
- **Grass:** In the mountain meadows of easy level there will be patches of grass to enhance the beauty.

Medium level

- **Retaining wall:** In the medium level the retaining wall on the balcony.
- **Door:** In the medium level simulation, there will be a door that opens to the balcony.
- **Benches:** In the medium level, some benches will be placed on the road that can be seen from the balcony of the building.
- **Plants:** The plants placed on the balcony of medium level.
- **Road:** The road that can be seen from the balcony in the medium level simulation.
- **Street lights:** In the medium level the road that players will be able to see from the high balcony of the building will have street lights.
- **Cars:** Multiple cars on the road that can be seen from the balcony in medium level.
- **Buildings:** In the medium level, multiple buildings that can be seen from the balcony.

Hard Level

- **SkyBox HDRP:** High quality skies that are appropriate for the desert canyon setting.
- **Shrubs and Bushes:** These will be scattered in the scene to show minimum vegetation.
- **Trees:** These will be scattered in the desert canyon environment.
- **Rocks:** Small sized rocks will be randomly placed in the user's path
- **Sand Textures:** These are textures for the ground user to walk on.

Conclusion

In conclusion, we have completed the implementation of the profile management module, and have created a working prototype for the entire desktop application. Additionally, we have prepared for iteration 2's Acrophobia simulations, by collecting assets, and by researching and collecting questionnaires related to acrophobia (fear of heights).

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